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PUBLIC REVIEW DRAFT: DUNE GIZMO – SC/DPS Interface Control Document

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Public review copy. This is not a controlled EDMS revision. Site addresses, credentials, and controlled installation records are intentionally omitted.

Summary

This ICD answers a specific integration question: how does the Ground Impedance Monitor (GIZMO) hand its measurements, alarms, and health information to DUNE Slow Controls and the Detector Protection System (SC/DPS)? At that boundary, every active GIZMO looks the same: one copper Gigabit Ethernet connection and one OPC UA server using the `urn:fnal:gizmo` information model. The Kria unit is the primary implementation and the legacy ZedBoard is the spare. Hardware differences remain visible through OPC UA quality and access metadata; they do not create a second SC design. GIZMO reports the ground-impedance alarm to SC/DPS. The response after that report is received is defined elsewhere.

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Distribution List

GIZMO system owner; DUNE Slow Controls and Detector Protection System; installation and integration; controls cybersecurity.

History of Changes

Revision	Date	Author	Change
0.10-public	2026-08-14	M. Arroyave	Dual-platform public review draft.
0.11-public	2026-08-17	M. Arroyave	Replaced the platform-by-platform design narrative with one concise interface; removed numbered GIZMO requirements and implementation/test history; identified the ZedBoard as the spare; clarified physical Ethernet, OPC UA, safety-reporting, and scope boundaries.
0.12-public	2026-08-17	M. Arroyave	Renamed the document as the GIZMO–SC/DPS ICD and revised the presentation so that context and operator meaning accompany the normative interface requirements.
0.13-public	2026-08-18	M. Arroyave	Synchronized the released OPC UA model baseline to 1.4.0 and clarified the boundary between GIZMO tag history and the separate SC–Computing database service.

Contents

1 Purpose and scope

GIZMO watches the electrical relationship between Detector Ground and Cavern/Building Ground. Its job at the controls boundary is straightforward: report what it measured, report whether its own alarm logic is active, and provide enough health information for SC/DPS to decide whether that report can be trusted. This applies during both construction and detector operation.

The normal interface boundary is:

$$\text{GIZMO measurement hardware} \longrightarrow \text{GIZMO controller} \xrightarrow{\text{1000BASE-T}} \text{controls network}$$
$$\text{controls network} \xrightarrow{\text{OPC UA}} \text{SC/Ignition and SC-managed DPS}$$

The ICD defines that network connection, the OPC UA contract, the meaning and quality of the data, the limited controls available to SC/DPS, and the tests used to accept the interface. It includes enough grounding and rack context to prevent the measurement from being misunderstood. It is not an electrical work procedure, and it does not decide what happens after SC/DPS receives an alarm. Notification, escalation, work-pause decisions, and protective actions belong to the applicable operating and protection procedures.

In this document, "shall" marks behavior that is required for interface acceptance. "Should" marks a recommendation. A value marked TBD is genuinely unresolved; it is not zero, disabled, or accepted by omission.

1.1 Applicable documents and precedence

Document	Why it matters at this interface
Released Far Detector grounding drawings and cable schedule	Authoritative physical grounding topology, conductor identity, CT orientation, and stimulus/return routing.
Current rack ORC; DUNE-doc 35392 as the development-rack reference	Rack configuration, power, installation, labeling, and safety authorization. The installed rack requires its own applicable approved ORC.
Maintained GIZMO runtime and generated `schema/gizmo-opcua-contract.json`	Normative OPC UA NodeIds, datatypes, units, ranges, access metadata, methods, and meanings.
DUNE-doc 1805 and 27482	Engineering reference for the legacy ZedBoard spare; not an alternate SC interface.
DUNE-doc 37315	DUNE Slow Controls architecture and integration direction.

Current DUNE SC–Computing ICD	Database-service boundary for Slow Controls: hosting, network access, backup, recovery, security, and support ownership. It does not change the GIZMO OPC UA producer contract.

The division of authority is important. Released drawings and the applicable ORC control the physical installation. The released generated OPC UA contract controls the machine interface. This ICD explains that contract for people. If this summary or an intake spreadsheet disagrees with either controlling source, GIZMO and SC/DPS resolve the discrepancy before the interface is used.

2 One interface, two hardware implementations

SC/DPS should not need a different integration simply because the electronics inside the rack changed. It connects to a GIZMO endpoint, checks the identity reported by that endpoint, and uses the common information model. Each active unit runs its own OPC UA server with its own device identity and endpoint. One unit never proxies, configures, starts, or stops the other.

Role	**Hardware**	**SC/DPS treatment**
Primary	AMD/Xilinx K26 SOM on a KR260 carrier	Full reference implementation of the common interface.
Spare	Legacy ZedBoard and preserved controller	Same namespace, NodeIds, datatypes, units, and meanings. Unsupported data remain visible with a specific non-good StatusCode; unsupported controls return `BadNotSupported`. A narrower hardware limit may reject a write without changing the common engineering metadata.

The hardware name remains useful for maintenance and may be shown on the display. It must not change the meaning of a tag. Dashboards, alarms, history, and operator workflow use the common model, so changing to the spare does not require a second SC/DPS tag design.

3 Physical interface

3.1 SC/DPS network connection

For SC/DPS, the physical handoff is one ordinary copper Ethernet link per active GIZMO. The details below make that statement testable and prevent a maintenance port from quietly becoming a second controls interface.

Item	**Interface**
Link	1000BASE-T Gigabit Ethernet in accordance with IEEE 802.3. The accepted operating state is 1 Gb/s, full duplex.
Medium	Four-pair balanced copper cable, Category 5e or better, installed to the site cabling standard. The complete channel, including patch cables, shall not exceed 100 m.
Connector	8P8C modular connector, commonly called RJ-45.
Negotiation	Auto-negotiation may be used; link speed and duplex shall be verified during commissioning.
Network	Site-assigned hostname, address, switch port, VLAN, route, time source, and firewall policy on the restricted controls network. These are installation records, not fixed in this public ICD.
Labeling	Both cable ends carry the device and switch-port identifiers. Label the rack with its equipment identifier and the applicable ORC number.

A second controller port, a maintenance laptop connection, USB, UART, JTAG, and local web pages are maintenance interfaces. They are not additional SC data or command paths.

3.2 Rack grounding and safety context

The network interface cannot be interpreted safely without knowing what the instrument is watching. Rack-door and endpoint labels therefore identify Detector Ground, Cavern/Building Ground, the monitored safety-ground conductor through the current transformer (CT), and the approved stimulus and return points. If the released installation adopts the site color convention, Cavern Ground is green and Detector Ground is green with a yellow stripe. Color is a visual aid, never the source of truth; the released drawing and durable text labels remain authoritative.

The detector is not simply left floating. An approved safety-ground path connects the two references through the saturable-inductor assembly. GIZMO observes that arrangement; it is not the safety ground.
Never lift the safety ground to clear a GIZMO alarm or restore a reading.

The released electrical design and ORC own the CT, stimulus, power, chassis bond, beacon/audio, and saturable-inductor connections. They are context for SC/DPS, not controls exposed by this ICD. SC/DPS must not alter or drive them through an undocumented path.

The rack door also carries a short, revision-controlled reference identifying the rack and device, the applicable ORC, both ground references, the meaning of the local alarm indications, and the safety statement above. That reference may point the reader to the controlled response procedure; it does not invent or summarize SC/DPS escalation logic.

4 Protocol

Once the Ethernet link is up, SC/DPS talks to GIZMO through OPC UA. Each active GIZMO shall expose one OPC UA server as its supported public machine interface; the client does not need to know how the controller moves data internally.

Item	Interface
Protocol	OPC UA Binary over UA TCP.
Endpoint	`opc.tcp://<assigned-host>:<assigned-port>`; TCP 4840 is the default. Each device has a distinct endpoint and application/device identity.
Namespace	`urn:fnal:gizmo`. Clients resolve the namespace URI at connection time and shall not hard-code its numeric namespace index.
Root object	`Objects/GIZMo`.
Contract	Released generated GIZMO OPC UA JSON schema, currently model 1.4.0. Stable string NodeIds, datatypes, ranks, units, ranges, access metadata, method signatures, quality, and timestamp meanings form the contract.
Client behavior	SC/DPS uses persistent OPC UA subscriptions. Repeatedly opening sessions or polling private device services is not supported.
Security	The endpoint resides on the restricted controls network. The approved site certificate, trust-list, role, and firewall policy shall be recorded at installation. An insecure policy requires an explicit approved exception and is not encryption.

Private device protocols, compatibility ports, SSH, HTTP dashboards, service managers, and raw register access may still be useful to a maintainer. They are not part of this interface, and an SC/DPS client must not depend on them.

5 Data and controls

5.1 Minimum SC/DPS view

The generated contract contains the complete address space. An operator does not need to see every node at once. The normal SC/DPS view should instead make three things immediately clear: which GIZMO is reporting, what it currently sees, and whether that report is trustworthy. The groups below are the minimum needed to answer those questions; exact NodeIds and metadata remain in the generated schema.

Group	Essential values	Purpose
Identity	DeviceId, product/platform, model version, runtime version, boot ID	Answers "Which instrument and software produced this report?"
Measurement	Sequence, source time, resistance, resistance range, active threshold, stimulus frequency, quality, diagnostic	Shows the measured state and whether samples are continuing to arrive.
Alarm	Active, Latched, Reason, LatchTime	Shows the device's authoritative ground-impedance/ground-short decision and persistent latch.
Configuration	ThresholdOhm, AveragesPerCalculation, LastCommandResult	Provides bounded configuration and command readback.
Calibration	State and active calibration identity/time	Distinguishes valid, unvalidated, and maintenance states.
Time	Current UTC and synchronization status	Establishes trustworthy source timestamps.
Health	Overall, measurement, network, services, firmware, and last update	Separates device/interface faults from a physical ground alarm.

Measurement and alarm values normally update once per second. Platform health changes more slowly and may update every 10 or 30 seconds. Each sample carries the time and sequence assigned at its source. SC/DPS preserves those values and does not fill a gap with invented intermediate samples.

5.2 Quality and value semantics

The number alone is never the whole report. Every value arrives with an OPC UA datatype, a StatusCode, and a source timestamp. Together they tell SC/DPS both what GIZMO measured and how much confidence to place in it. Unknown or unavailable numeric data must not masquerade as zero. Before the first sample, the value reports a waiting or bad state. If the source is later lost, the last value may remain visible, but its status becomes uncertain or bad so that it cannot look current.

Resistance illustrates why those fields must be read together. `ResistanceOhm` is numeric only when `ResistanceRange=InRange`. Above the validated presentation range, GIZMO reports `ResistanceRange=OutOfRange`; the display calls this `HIGHZ`, and the resistance itself is non-numeric. A good-quality HIGH Z report is a valid range state. It is neither a sensor fault nor an invented 500- or 999-ohm measurement.

`Alarm.Active` comes from the device logic that drives the physical alarm output. That Boolean is the alarm decision SC/DPS uses. Recreating the decision with a separate resistance-versus-threshold comparison could disagree with the instrument and is therefore not allowed. The report says that the configured alarm condition exists; it cannot locate or diagnose the conductive path.

5.3 Controls boundary

Most information crosses this interface from GIZMO to SC/DPS. The few controls that travel in the other direction are deliberately bounded and must always return an observable result.

Control	**Class**	**Rule**
ThresholdOhm	Bounded read/write	Authorized operator setting; verify the live measurement readback after writing. The spare may enforce a narrower accepted range and return `BadOutOfRange`.
AveragesPerCalculation	Bounded read/write when supported	Verify readback and a new measurement sequence. An unsupported spare write returns `BadNotSupported`.
ClearLatch	Operator method when supported	Clears only the persistent latch; it shall not suppress an active physical alarm.
Calibration, ADC capture, normalization, and manual time operations	Maintenance-only	Disabled in the normal SC view until authorization, readback, failure cleanup, and commissioning are approved.

Every write or method call is typed, bounded, role-authorized, serialized, logged, and followed by a result or readback. A successful method return means that GIZMO accepted the request; the resulting measurement sequence or state change is what demonstrates completion. Shell commands, service commands, reboots, file paths, FPGA/register writes, relay manipulation, and raw calibration-resistor selection are maintenance actions, not SC/DPS controls.

6 How SC/DPS uses the report

6.1 Display, alarms, and history

An operator first needs to know whether there is a ground alarm, but a clear alarm is meaningful only when the underlying report is current and valid. The main display therefore shows device identity, endpoint health, data age, StatusCode/quality, resistance or HIGH Z, threshold, alarm/latch/reason, and calibration state together.

SC/DPS presents the following as distinct conditions rather than collapsing them into a single generic GIZMO alarm:

- the physical ground-impedance alarm reported by `GIZMo.Alarm.Active`;
- communication loss or a stale timestamp/sequence;
- non-good measurement quality or invalid calibration;
- unexpected device/model identity; and
- a failed or rejected command.

The stale-data limit is selected and approved during commissioning. Crossing that limit raises a monitoring-coverage alarm. It does not turn the last value into a good report, and it must not be mistaken for a clear ground state. The operating procedure—not this ICD—decides what activity may continue while coverage is unavailable.

Ignition subscribes directly to the GIZMO OPC UA endpoint. Its history retains the device identity, source timestamp, StatusCode, range state, alarm transitions, and command audit throughout construction and operation. When the ground-impedance report reaches the SC-managed DPS integration, it still carries the quality and source timestamp supplied by GIZMO.

The database behind Ignition is not a GIZMO device-interface requirement. The selected long-term direction is PostgreSQL, while hosting, network access, retention, backup, recovery, security, and service ownership follow the SC–Computing ICD and site operations. An interim test database does not define the production service.

That is the end of this interface: SC/DPS has received, quality-checked, displayed, alarmed, and recorded the report. Notification, escalation, work-control, and automatic or manual protective response begin on the other side of that boundary and are defined elsewhere.

7 Roles and responsibilities

Responsibility follows the report: GIZMO owns it at the source; SC/DPS owns its secure delivery, presentation, freshness supervision, and recording. This allocation stops at the reporting boundary. The downstream response is defined elsewhere.

The GIZMO system owner owns the instrument, common OPC UA contract, device software, value meanings, calibration authority, and safe local alarm behavior. The owner supplies the device identity, data, quality, and source timestamps.

SC/DPS owns the secure OPC UA integration, operator display, supervisory and stale-data alarms, Ignition history, command audit, and delivery of the GIZMO report to DPS. Computing service responsibilities remain allocated by the separate SC–Computing ICD.

Together, GIZMO and SC/DPS commission the endpoint, map the signals, accept the interface, and control later changes.

8 Interface acceptance

Acceptance follows a report from the device to the operator display and DPS handoff, then checks the few controls in the reverse direction. Before the interface is released, GIZMO and SC/DPS must jointly complete these checks:

1. verify the rack ID and applicable ORC label, cable labels, assigned switch port, 1 Gb/s full-duplex link, VLAN/addressing, route, and time source;
2. connect with a generic OPC UA client, resolve `urn:fnal:gizmo`, and verify device/application identity;
3. run the versioned schema conformance test for required NodeIds, datatypes, units, access, StatusCodes, and source timestamps;
4. verify once-per-second measurement sequence/time progression and the commissioned stale-data alarm;
5. with an approved test input, verify finite resistance, HIGH Z, the device alarm state, latch, reason, bad or unavailable data, disconnect, restart, and recovery without fabricated values;
6. verify that Ignition and the SC-managed DPS mapping use `Alarm.Active`, preserve source quality and time in history, and record command results/readbacks;
7. reject anonymous or unauthorized writes and every out-of-range or unsupported operation explicitly; and
8. demonstrate that loss of OPC UA or Ignition history does not stop local measurement or the physical alarm.

The spare passes the same interface tests before it is declared available. A successful interface test demonstrates that data and controls cross the boundary correctly. It does not establish electrical authorization, metrology, calibration validity, or approval of any response downstream of SC/DPS.

9 Items requiring controlled approval

The interface can be described publicly without publishing site configuration. The decisions below are therefore completed in controlled records before deployment; none may be filled in by assumption or inherited silently from a prototype.

Decision	**Owner**
Assign the EDMS number, reviewers, approvers, and controlled distribution.	GIZMO + SC/DPS
Assign the underground rack ORC and place its number on the rack label.	Installation + electrical safety
Approve production endpoint, switch/VLAN, addressing, certificates, roles, and firewall rules.	SC + cybersecurity
Approve site threshold, stale-data limit, alarm priority, and SC/DPS signal mapping.	GIZMO + SC/DPS
Approve the rack-door reference content and revision control.	GIZMO + SC/DPS + electrical safety
Complete common-interface and physical-display acceptance of the ZedBoard spare.	GIZMO + SC/DPS